

CS 6371 - Assignment 4

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1. (a) Denotationally,

$$C\llbracket c_1; c_2 \rrbracket = \{(\sigma, C\llbracket c_2 \rrbracket(C\llbracket c_1 \rrbracket(\sigma))) \mid \sigma \in \Sigma\} \quad (1)$$

By definition, $C\llbracket c_1; c_2 \rrbracket$ is confidentiality-preserving if the following property holds:

$$\begin{aligned} \forall \sigma_1, \sigma_2 \in \Sigma. ((C\llbracket c_1 \rrbracket \sigma_1) \mid_L = (C\llbracket c_1 \rrbracket \sigma_2) \mid_L) \\ \implies (C\llbracket c_2 \rrbracket(C\llbracket c_1 \rrbracket \sigma_1) \mid_L = C\llbracket c_2 \rrbracket(C\llbracket c_1 \rrbracket \sigma_2) \mid_L) \end{aligned} \quad (2)$$

If we assume c_1 and c_2 are confidentiality-preserving:

$$\forall \sigma_1, \sigma_2 \in \Sigma. ((\sigma_1 \mid_L = \sigma_2 \mid_L) \implies ((C\llbracket c_1 \rrbracket \sigma_1) \mid_L = (C\llbracket c_1 \rrbracket \sigma_2) \mid_L)) \quad (3)$$

$$\forall \sigma_3, \sigma_4 \in \Sigma. ((\sigma_3 \mid_L = \sigma_4 \mid_L) \implies ((C\llbracket c_2 \rrbracket \sigma_3) \mid_L = (C\llbracket c_2 \rrbracket \sigma_4) \mid_L)) \quad (4)$$

We apply assumption 4 to assumption 3, with $\sigma_3 = C\llbracket c_1 \rrbracket \sigma_1$ and $\sigma_4 = C\llbracket c_1 \rrbracket \sigma_2$ to conclude:

$$\begin{aligned} \forall \sigma_1, \sigma_2 \in \Sigma. ((C\llbracket c_1 \rrbracket \sigma_1) \mid_L = (C\llbracket c_1 \rrbracket \sigma_2) \mid_L) \\ \implies (C\llbracket c_2 \rrbracket(C\llbracket c_1 \rrbracket \sigma_1) \mid_L = C\llbracket c_2 \rrbracket(C\llbracket c_1 \rrbracket \sigma_2) \mid_L) \end{aligned} \quad (5)$$

$(\sigma_1 \mid_L = \sigma_2 \mid_L)$ by assumption. (The confidentiality-preserving property says nothing about states whose low-confidentiality variables don't match.)

Therefore, $C\llbracket c_1; c_2 \rrbracket$ is confidentiality-preserving. $\square$

- (b) Consider variables $x, y \in \sigma^{\leftarrow}$ such that $y \in \sigma \mid_L^{\leftarrow}$ and $x \notin \sigma \mid_L^{\leftarrow}$

Given these variables, the following program is not confidentiality-preserving:

$$c = \text{if } x \leq 1 \text{ then } y := 1 \text{ else } y := 0 \quad (6)$$

For example, consider the initial states $\sigma_1 = \{(x, 0), (y, 0)\}$ and $\sigma_2 = \{(x, 2), (y, 0)\}$. In this case,

$$\begin{aligned} C\llbracket c \rrbracket(\sigma_1) &= \{(x, 0), (y, 1)\} \\ C\llbracket c \rrbracket(\sigma_2) &= \{(x, 2), (y, 0)\} \end{aligned} \quad (7)$$

Since $\sigma_1 \mid_L = \sigma_2 \mid_L = \{(y, 0)\}$, but $C\llbracket c \rrbracket(\sigma_1) \mid_L \neq C\llbracket c \rrbracket(\sigma_2) \mid_L$, the program is not confidentiality-preserving.

2. (a)

$$F(g) = \lambda x. (x \leq 2 \rightarrow x - 1 \mid x > 2 \text{ and } x \text{ is odd} \rightarrow (x - 3) \cdot g(x - 1) \cdot g(x + 1) \\ \mid x > 2 \text{ and } x \text{ is even} \rightarrow g(x/2) + \text{sgn}(g(x/2))) \quad (8)$$

(b)

$$h(x) = \begin{cases} \log_2(x) & \text{if } \log_2(x) \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

(c) *Proof.* Define property P by $P(g) \equiv \forall x \in g^\leftarrow. g(x) = h(x)$. We wish to prove $P(f)$. Define functional F as above, and observe that $\text{fix}(F) = f$ by the definition of recursion. Thus, to prove $P(f)$ it suffices to prove $P(\text{fix}(F))$ by fixed-point induction.

Base Case: $P(\perp)$ holds vacuously.

Inductive Hypothesis: Assume that $P(g)$ holds for some arbitrary function g . That is, assume that $\forall x_0 \in g^\leftarrow, g(x_0) = h(x_0)$.

Inductive Case: We will prove that $P(F(g))$ holds. Let $x \in F(g)^\leftarrow$ be given. Looking at the definition of F , there are three cases to consider:

Case 1: Suppose $x \leq 2$. Then by definition of F ,

$$\begin{aligned} F(g)(x) &= x - 1 \\ F(g)(1) &= 0 = \log_2(1) = h(1) \\ F(g)(2) &= 1 = \log_2(2) = h(2) \end{aligned} \quad (10)$$

Case 2: Suppose $x > 2$ and x is odd. Then by definition of F ,

$$F(g)(x) = (x - 3) \cdot g(x - 1) \cdot g(x + 1) \quad (11)$$

In this case, $x - 1$ and $x + 1$ will be even numbers with difference 2. If we start from 1, every $\log_2$ will be double the previous (1, 2, 4, 8, 16, ...). Therefore, the only integers with difference 2 for which $\log_2$ is an integer are 2 and 4. In this case, $x = 3$ and $x - 3 = 0$, so $F(g)(3) = 0 = h(3)$. For every other case, $\log_2$ of either $x - 1$ or $x + 1$ will not be an integer. Therefore, applying the inductive hypothesis would yield 0 for at least one of them and $F(g)(x) = 0 = h(x)$.

Case 3: Suppose $x > 2$ and x is even. Then by definition of F ,

$$F(g)(x) = g(x/2) + \text{sgn}(g(x/2)) \quad (12)$$

There are two subcases to consider:

Subcase 1: Suppose $\log_2(x/2)$ is not an integer. By inductive hypothesis, $g(x/2) = 0$; thus $F(g)(x) = 0 + \text{sgn}(0) = 0 = h(x)$.

Subcase 2: Suppose $\log_2(x/2)$ is an integer. By inductive hypothesis, $F(g)(x) = \log_2(x/2) + \text{sgn}(\log_2(x/2))$. Clearly $\text{sgn}(\log_2(x/2)) = 1$ since $x > 2$. Therefore, $F(g)(x) = \log_2(x/2) + 1 = \log_2(x) = h(x)$. $\square$

3. (a)

$$\Gamma(f) = \{(\sigma, \sigma) \mid 1 > \sigma(x)\} \cup \{(\sigma, f(\sigma[y \mapsto 1 - \sigma(y)][x \mapsto \sigma(x) - 1])) \mid 1 \leq \sigma(x)\} \quad (13)$$

- (b) *Proof.* Property $P'(f)$ implies $P(f)$ because P is merely the special case of P' when $\sigma(y) = 0$. Proving $P'(C\llbracket c \rrbracket)$ therefore suffices to prove $P(C\llbracket c \rrbracket)$. We will prove $P'(C\llbracket c \rrbracket)$ by fixed-point induction over Γ . By definition of C , $C\llbracket c \rrbracket = fix(\Gamma)$ where Γ is defined in part a.

We can therefore prove $P'(C\llbracket c \rrbracket)$ by fixed point induction over Γ .

Base Case: $P'(\perp)$ holds vacuously.

Inductive Hypothesis: Let $g : \Sigma \rightarrow \Sigma$ be an arbitrary function satisfying P' . That is, assume

$$\forall(\sigma_0, \sigma'_0) \in g, \left((\sigma_0(x) \geq 0 \wedge \sigma_0(y) \in \{0, 1\}) \implies \sigma'_0(y) = \begin{cases} \sigma_0(y) & \text{if } \sigma_0(x) \text{ is even} \\ 1 - \sigma_0(y) & \text{if } \sigma_0(x) \text{ is odd} \end{cases} \right) \quad (14)$$

Inductive Case: Let $(\sigma, \sigma') \in \Gamma(g)$ be given. Assume $\sigma(x) \geq 0$ and $\sigma(y) \in \{0, 1\}$.

Case 1: Assume $1 > \sigma(x)$. By definition of Γ , $\sigma' = \sigma$. Since $0 \leq \sigma(x) < 1$, we infer that $\sigma(x) = 0$. Zero is even and $\sigma'(y) = \sigma(y)$, therefore the property holds.

Case 2: Assume $1 \leq \sigma(x)$. By definition of Γ , $\sigma' = g(\sigma_2)$ where $\sigma_2 = \sigma[y \mapsto 1 - \sigma(y)][x \mapsto \sigma(x) - 1]$. We apply the inductive hypothesis with $\sigma_0 = \sigma_2$ and $\sigma'_0 = \sigma'$:

- $\sigma_2(x) \geq 0$ because $\sigma_2(x) = \sigma(x) - 1$ and $\sigma(x) \geq 1$
- $\sigma_2(y) \in \{0, 1\}$ because $\sigma(y) \in \{0, 1\}$ and $\sigma_2(y) = 1 - \sigma(y)$

From the inductive hypothesis, we infer that

$$\begin{aligned} \sigma'(y) &= \begin{cases} \sigma_2(y) & \text{if } \sigma_2(x) \text{ is even} \\ 1 - \sigma_2(y) & \text{if } \sigma_2(x) \text{ is odd} \end{cases} \\ &= \begin{cases} 1 - \sigma(y) & \text{if } \sigma_2(x) \text{ is even} \\ 1 - (1 - \sigma(y)) & \text{if } \sigma_2(x) \text{ is odd} \end{cases} \\ &= \begin{cases} 1 - \sigma(y) & \text{if } \sigma_2(x) \text{ is even} \\ \sigma(y) & \text{if } \sigma_2(x) \text{ is odd} \end{cases} \end{aligned} \quad (15)$$

Since $\sigma_2(x) = \sigma(x) - 1$, $\sigma_2(x)$ is even when $\sigma(x)$ is odd, and vice versa. Thus,

$$\sigma'(y) = \begin{cases} \sigma(y) & \text{if } \sigma(x) \text{ is even} \\ 1 - \sigma(y) & \text{if } \sigma(x) \text{ is odd} \end{cases} \quad (16)$$

Therefore, $P'(C\llbracket c \rrbracket)$ holds. $\square$